DS-UNet: Dual-Stream U-Net for Oil Spill Detection of SAR Image

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IEEE Geoscience and Remote Sensing Letters, 2023

Abstract

This paper proposes a dual-stream U-Net (DS-UNet) architecture for oil spill detection in synthetic aperture radar (SAR) images. Unlike traditional U-Net- or Transformer-based models that rely on single-scale feature representations, the proposed DS-UNet integrates edge, local, and global features through a dual-stream design. The model consists of an edge feature extraction module and an interscale alignment module, enabling robust feature fusion under speckle noise and complex ocean conditions. Experiments on PALSAR and Sentinel datasets demonstrate that DS-UNet outperforms existing state-of-the-art segmentation methods.

1 Introduction

Oil spills are among the most harmful forms of marine pollution, causing severe damage to marine ecosystems. Timely and accurate detection of oil spills is therefore critical for environmental monitoring and disaster response. Oil spill detection from SAR imagery is typically formulated as a pixel-level semantic segmentation task.

Traditional unsupervised methods, such as thresholding and clustering, rely solely on pixel-level information and ignore spatial context. Convolutional neural networks (CNNs), particularly U-Net-based architectures, have achieved significant success in segmentation tasks by leveraging local spatial features. However, CNNs are limited in modeling long-range dependencies.

Recently, Transformer-based models have been introduced to capture global contextual information. Hybrid CNN–Transformer models such as TransUNet combine local and global features but primarily focus on the deepest feature map, potentially neglecting information from other feature scales. Moreover, existing methods often overlook the influence of speckle noise and edge information in SAR images.

To address these limitations, this paper proposes DS-UNet, a dual-stream architecture that explicitly integrates edge features, local features, and global features across multiple scales.

2 Proposed Method

The proposed DS-UNet takes a SAR image as input and produces a pixel-wise classification map indicating oil spill and non-oil-spill regions. The overall architecture follows a U-shaped encoder–decoder structure and consists of four key components: an edge extraction block, a CNN–Transformer hybrid encoder, an interscale alignment module, and a decoder.

The dual-stream design allows information to propagate between local and global feature representations while preserving edge details critical for accurate boundary localization.

3 Edge Feature Extraction Module

SAR images are prone to speckle noise, and oil spill boundaries often exhibit weak contrast. To enhance boundary awareness, an edge extraction block is introduced at the input stage.

The Sobel operator is employed to extract horizontal and vertical edge responses. These edge features emphasize intensity transitions and suppress isolated noisy regions. The extracted edge map is fused with the original SAR image through convolutional layers and normalization operations, producing an edge-enhanced feature map.

This module guides subsequent feature extraction stages to focus on meaningful boundary information and improves robustness against speckle noise.

4 CNN–Transformer Hybrid Encoder

The encoder is designed to extract both local and global features. A modified ResNet50 backbone is used to extract multiscale local features at resolutions of one-half, one-quarter, one-eighth, and one-sixteenth of the original image size.

To capture global dependencies, the deepest feature map is fed into a Transformer layer. Position embeddings and multihead self-attention enable modeling of long-range relationships across the image. This hybrid design leverages the strengths of both CNNs and Transformers.

5 Interscale Alignment Module

The interscale alignment module is the core innovation of DS-UNet. It fuses high-resolution local feature maps with low-resolution global feature maps generated by the Transformer.

Cross-attention mechanisms are used to align and integrate features from different scales. This bidirectional information exchange allows the model to preserve fine spatial details while incorporating global semantic context.

The output of this module is a set of hybrid feature maps that contain both local and global information, which are then passed to the decoder.

6 Decoder

The decoder restores the spatial resolution of the feature maps through a sequence of upsampling and convolutional operations. Skip connections are employed to combine interscale-aligned features with corresponding encoder features.

This symmetric U-shaped design ensures effective recovery of spatial details and accurate localization of oil spill boundaries.

7 Experimental Setup

Experiments are conducted on two real-world datasets: the PALSAR dataset and the Sentinel dataset. Both datasets consist of SAR images containing oil spill events under diverse environmental conditions.

Data augmentation techniques such as random rotation and flipping are applied to improve generalization. The model is trained using stochastic gradient descent with carefully selected hyperparameters. A mixed loss function combining Dice loss and cross-entropy loss is employed to balance region accuracy and boundary precision.

Evaluation metrics include Dice Similarity Coefficient (DSC), Hausdorff Distance (HD), F1 score, and inference time.

8 Methodology

The overall methodology of DS-UNet can be summarized as follows:

1. Apply Sobel-based edge extraction to enhance boundary features.
2. Extract multiscale local features using a CNN encoder.
3. Capture global contextual information using a Transformer layer.
4. Fuse local and global features through interscale alignment.
5. Restore spatial resolution using a U-Net-style decoder.
6. Optimize the network using a combined Dice and cross-entropy loss.

This integrated approach enables robust oil spill detection under speckle noise and complex ocean conditions.

9 Performance Analysis

DS-UNet is compared with eight state-of-the-art segmentation models, including U-Net, FCN, TransUNet, R2U-Net, AttnUNet, NestedUNet, Swin-Unet, and SegNet.

Experimental results show that DS-UNet achieves the highest DSC and F1 scores on both PALSAR and Sentinel datasets while maintaining competitive inference time. The model significantly improves boundary accuracy, as reflected by lower Hausdorff Distance values.

Ablation studies demonstrate that removing the edge extraction block degrades performance, confirming its importance. Visual comparisons further reveal that DS-UNet produces sharper boundaries and fewer false negatives compared to competing methods.

10 Conclusion

This paper introduces DS-UNet, a dual-stream U-Net architecture for oil spill detection in SAR images. By integrating edge features, local features, and global features across multiple scales, the proposed method effectively addresses the challenges of speckle noise and complex oil spill morphology.

Extensive experiments on real SAR datasets confirm that DS-UNet achieves state-of-the-art performance. The proposed framework provides a promising solution for practical oil spill monitoring applications.